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Assignment

Bachelor of Computer Applications
Semester III

Database Management System
Topic: Database Design for a Real-World
Application

BY

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DBMS ASSIGNMENT

Database Design for a Real-World Application

1. Name and Purpose of the Application

Application chosen: Hospital Management System (HMS)

A Hospital Management System is used to digitize and streamline the day-to-day operations of a hospital. It manages patient records, doctor and department information, appointment scheduling, in-patient admissions and room allocation, prescriptions and pharmacy stock, diagnostic lab tests, and billing. The goal of the underlying database is to store all of this information in a structured, non-redundant way, and to make it possible to answer real operational questions quickly — for example, which rooms are free, what medicines a patient was prescribed, or how much a patient currently owes the hospital.

2. List of Required Tables

- Department
- Doctor
- Patient
- Appointment
- Room
- Admission
- Prescription
- Medicine
- Prescription_Detail
- Lab_Test
- Bill

3 & 5. Tables, Columns, and Purpose of Each Table

Each table below is introduced with a short explanation of its purpose, followed by its columns, data types, and key roles.

1. Department

Stores information about each hospital department (e.g. Cardiology, Orthopedics, Radiology). Every doctor belongs to exactly one department, so this table lets the hospital organize staff, resources, and location by specialty.

Column	Data Type	Key	Description
department_id	INT	PK	Unique identifier for the department
department_name	VARCHAR(50)		Name of the department, e.g. 'Cardiology'
floor_no	INT		Floor on which the department is located

Column	Data Type	Key	Description
phone_no	VARCHAR(15)		Contact number for the department desk
head_doctor_id	INT	FK -> Doctor	Doctor who heads the department

2. Doctor

Stores details of every doctor working in the hospital, including their specialization and the department they belong to. Referenced by Appointment, Admission, Prescription, and Lab_Test to identify which doctor is responsible for each activity.

Column	Data Type	Key	Description
doctor_id	INT	PK	Unique identifier for the doctor
name	VARCHAR(60)		Full name of the doctor
specialization	VARCHAR(50)		Medical specialization, e.g. 'Neurologist'
department_id	INT	FK -> Department	Department the doctor belongs to
phone_no	VARCHAR(15)		Doctor's contact number
email	VARCHAR(60)		Doctor's official email address
experience_years	INT		Years of professional experience
consultation_fee	DECIMAL(8,2)		Standard fee charged per consultation

3. Patient

Stores personal and contact details of every patient registered with the hospital. This is the central table that Appointment, Admission, Prescription, Lab_Test, and Bill all link back to.

Column	Data Type	Key	Description
patient_id	INT	PK	Unique identifier for the patient
name	VARCHAR(60)		Full name of the patient
dob	DATE		Date of birth, used to compute age
gender	VARCHAR(10)		Gender of the patient
phone_no	VARCHAR(15)		Patient's contact number
address	VARCHAR(150)		Residential address
blood_group	VARCHAR(5)		Blood group, e.g. 'O+'
emergency_contact	VARCHAR(15)		Phone number of an emergency contact person

4. Appointment

Records every consultation booked by a patient with a doctor. This is the linking (junction) table that resolves the many-to-many relationship between Patient and Doctor: one patient can see many doctors over time, and one doctor sees many patients.

Column	Data Type	Key	Description
appointment_id	INT	<i>PK</i>	Unique identifier for the appointment
patient_id	INT	<i>FK -> Patient</i>	Patient who booked the appointment
doctor_id	INT	<i>FK -> Doctor</i>	Doctor the appointment is booked with
appointment_date	DATE		Date of the consultation
appointment_time	TIME		Time slot of the consultation
status	VARCHAR(15)		'Scheduled', 'Completed', or 'Cancelled'
reason	VARCHAR(150)		Brief reason for the visit

5. Room

Stores details of every room/bed available in the hospital (General Ward, ICU, Private, etc.), along with its daily charge and current occupancy status. Used by Admission to allocate a bed to an in-patient.

Column	Data Type	Key	Description
room_id	INT	<i>PK</i>	Unique identifier for the room
room_number	VARCHAR(10)		Physical room/bed number
room_type	VARCHAR(20)		'General', 'Private', 'ICU', etc.
floor_no	INT		Floor on which the room is located
daily_charge	DECIMAL(8,2)		Charge per day for occupying the room
status	VARCHAR(15)		'Available' or 'Occupied'

6. Admission

Records every in-patient stay: which patient was admitted, to which room, under which doctor's care, and for how long. Separated from Appointment because admissions involve a room, a duration, and a discharge process that a simple consultation does not.

Column	Data Type	Key	Description
admission_id	INT	<i>PK</i>	Unique identifier for the admission record
patient_id	INT	<i>FK -> Patient</i>	Patient being admitted
room_id	INT	<i>FK -> Room</i>	Room/bed allocated to the patient
doctor_id	INT	<i>FK -> Doctor</i>	Doctor supervising the admission
admission_date	DATE		Date the patient was admitted
discharge_date	DATE		Date the patient was discharged (NULL if still admitted)

Column	Data Type	Key	Description
admission_reason	VARCHAR(150)		Reason for hospitalization
status	VARCHAR(15)		'Admitted' or 'Discharged'

7. Prescription

Records that a doctor issued a prescription to a patient, typically following an appointment. Acts as the header record; the actual medicines prescribed are stored in Prescription_Detail.

Column	Data Type	Key	Description
prescription_id	INT	<i>PK</i>	Unique identifier for the prescription
appointment_id	INT	<i>FK -> Appointment</i>	Appointment during which it was issued
patient_id	INT	<i>FK -> Patient</i>	Patient the prescription is for
doctor_id	INT	<i>FK -> Doctor</i>	Doctor who issued the prescription
date_issued	DATE		Date the prescription was written
notes	VARCHAR(200)		Additional instructions from the doctor

8. Medicine

Acts as a catalog of medicines available in the hospital pharmacy, with price and stock information. Enables inventory tracking and is reused across many prescriptions.

Column	Data Type	Key	Description
medicine_id	INT	<i>PK</i>	Unique identifier for the medicine
medicine_name	VARCHAR(60)		Name of the medicine
manufacturer	VARCHAR(60)		Company that manufactures the medicine
unit_price	DECIMAL(8,2)		Price per unit/strip
stock_quantity	INT		Units currently available in the pharmacy

9. Prescription_Detail

Junction table resolving the many-to-many relationship between Prescription and Medicine: a single prescription usually lists several medicines, and the same medicine appears in many prescriptions. Each row is one medicine line-item within one prescription.

Column	Data Type	Key	Description
prescription_detail_id	INT	<i>PK</i>	Unique identifier for the line item
prescription_id	INT	<i>FK -> Prescription</i>	Prescription this line item belongs to

Column	Data Type	Key	Description
medicine_id	INT	<i>FK -> Medicine</i>	Medicine being prescribed
dosage	VARCHAR(30)		e.g. '1 tablet twice daily'
duration_days	INT		Number of days the medicine should be taken
quantity	INT		Total units to be dispensed

10. Lab_Test

Records diagnostic tests (blood test, X-ray, MRI, etc.) ordered for a patient by a doctor, along with the result and status. Kept separate from Prescription since a lab test is ordered and resulted independently of medicine dispensing.

Column	Data Type	Key	Description
test_id	INT	<i>PK</i>	Unique identifier for the lab test
patient_id	INT	<i>FK -> Patient</i>	Patient undergoing the test
doctor_id	INT	<i>FK -> Doctor</i>	Doctor who ordered the test
test_name	VARCHAR(60)		e.g. 'Complete Blood Count', 'Chest X-ray'
test_date	DATE		Date the test was conducted
result	VARCHAR(200)		Test findings/result summary
status	VARCHAR(15)		'Pending' or 'Completed'

11. Bill

Consolidates charges for a patient's visit — consultation fees, room charges, medicines, and tests — into a single payable bill. Links optionally to an Appointment (out-patient visit) or an Admission (in-patient stay).

Column	Data Type	Key	Description
bill_id	INT	<i>PK</i>	Unique identifier for the bill
patient_id	INT	<i>FK -> Patient</i>	Patient being billed
appointment_id	INT	<i>FK -> Appointment (nullable)</i>	Related appointment, if the bill is for an OPD visit
admission_id	INT	<i>FK -> Admission (nullable)</i>	Related admission, if the bill is for an in-patient stay
bill_date	DATE		Date the bill was generated
total_amount	DECIMAL(10,2)		Total amount payable
payment_status	VARCHAR(15)		'Paid', 'Pending', or 'Partially Paid'
payment_mode	VARCHAR(20)		'Cash', 'Card', 'UPI', 'Insurance', etc.

4. Relationships Between the Tables

The following relationships connect the tables described above:

- Department (1) — (M) Doctor: one department employs many doctors; each doctor belongs to exactly one department.
- Doctor (1) — (M) Appointment and Patient (1) — (M) Appointment: Appointment is the junction table that resolves the many-to-many relationship between Doctor and Patient — one doctor has many appointments, one patient books many appointments.
- Patient (1) — (M) Admission, Doctor (1) — (M) Admission, Room (1) — (M) Admission: an admission always belongs to one patient, is supervised by one doctor, and occupies one room; each of these can be linked to many admissions over time.
- Appointment (1) — (M) Prescription and Patient/Doctor (1) — (M) Prescription: a prescription is normally issued during one appointment, for one patient, by one doctor.
- Prescription (1) — (M) Prescription_Detail and Medicine (1) — (M) Prescription_Detail: Prescription_Detail is the junction table resolving the many-to-many relationship between Prescription and Medicine.
- Patient (1) — (M) Lab_Test and Doctor (1) — (M) Lab_Test: a lab test belongs to one patient and is ordered by one doctor; a patient/doctor can be linked to many lab tests.
- Patient (1) — (M) Bill: a patient can have many bills over time. Bill optionally links to one Appointment (OPD visit) or one Admission (in-patient stay).

Summary

Patient and Doctor form the two core entities around which the system revolves. Every other table either describes them further (Department for Doctor), records an interaction between them (Appointment, Admission, Prescription, Lab_Test), or supports those interactions (Room, Medicine, Prescription_Detail, Bill). Two junction tables — Appointment and Prescription_Detail — resolve the natural many-to-many relationships in the domain (a doctor sees many patients and vice versa; a prescription contains many medicines and a medicine appears in many prescriptions), which keeps the design free of redundant data while still capturing the full picture of hospital operations.